

Nonhomogeneous Equations & Undetermined Coefficients



Complementary function and particular solution

- 1 For $y'' - 3y' + 2y = 4e^{3x}$, find the complementary function y_c .
 - 2 Explain why the general solution of a nonhomogeneous linear equation is $y = y_c + y_p$.
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Choosing a trial form (no resonance)

- 3 Find a particular solution of $y'' - 3y' + 2y = 4x$.
 - 4 Find a particular solution of $y'' + y' - 2y = 6e^{3x}$ (note $r = 3$ is not a root of $r^2 + r - 2 = 0$).
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The modification rule (resonance)

- 5 Find a particular solution of $y'' - 3y' + 2y = e^{2x}$. (Note $r = 2$ IS a root of $r^2 - 3r + 2$.)
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Combining forcing terms

- 6 Find a particular solution of $y'' - y = e^x + x$. (Roots of $r^2 - 1 = 0$ are $r = \pm 1$, so e^x resonates but x does not.)
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Application: a driven oscillator

- 7 A forced oscillator satisfies $y'' + 4y = \sin 3t$ (forcing frequency $3 \neq$ natural frequency 2 — no resonance). Find a particular solution.
- 8 Now solve $y'' + 4y = \sin 2t$, where the forcing frequency EQUALS the natural frequency (resonance). Find a particular solution.